Massive Univariate BCFW Slices for Kinematic Reconstruction

October 28, 2026

Abstract

Finite-field functional reconstruction of multi-loop integral coefficients is cheapest along a *univariate slice* $\vec{s}(t) = \vec{a} + \vec{b}\,t$: a line in the space of Mandelstam invariants along which every invariant is an exact degree-one polynomial in t. For six or more external legs, momentum conservation and four-dimensionality force a nontrivial algebraic relation among the invariants (the *Gram constraint*), and a line that satisfies it exactly for every t is not obvious to write down by hand. This note explains, from first principles, how to build such a line out of spinor-helicity data so that the Gram constraint holds *automatically*, works through a fully explicit massless example, and then shows exactly what has to change – and, just as importantly, what does *not* – when two of the external legs are massive, as in $t\bar{t}b\bar{b}$-type kinematics.

1 The problem

Consider n external momenta $p_1, \ldots, p_n$ satisfying momentum conservation $\sum_i p_i = 0$ and living in four-dimensional Minkowski space. Define the Gram determinant of a set of vectors $u_1, \ldots, u_k$ by

$$G(u_1, \ldots, u_k) \;\coloneqq\; \det\left(2\, u_i \cdot u_j\right)_{i,j=1}^{k}. \tag{1}$$

Proposition 1 (The n-point Gram constraint)**.** If $v_1, \ldots, v_5$ are any five vectors in a space of dimension ≤ 4, then $G(v_1, \ldots, v_5) = 0$.

Proof. Five vectors in a ≤ 4-dimensional space are linearly dependent, so the 5×5 Gram matrix $(2v_i \cdot v_j)$ has rank ≤ 4 and vanishing determinant. □

Applying this to any five of the n external momenta gives $G(p_{i_1}, \ldots, p_{i_5}) = 0$: a single nontrivial polynomial relation among the Mandelstam invariants once $n \geq 6$. Two features of Proposition 1 are worth stressing because the rest of this note hinges on them:

- **It is pure linear algebra.** Nothing about masslessness, on-shellness, or which invariants you call independent enters the proof – only that the vectors live in ≤ 4 dimensions. Whatever makes legs massive cannot break it, and nothing needs to be done specially to *enforce* it on a genuine 4D configuration: it is automatic.
- **It is a relation, not a parametrization.** Knowing $G = 0$ tells you the invariants are not all independent, but solving it directly for one of them (say s_{16}) requires a square root of the discriminant. A line built by solving the constraint this way is *quadratic*, not linear, in its parameter – exactly the obstruction we want to avoid.

The strategy below sidesteps the square root entirely: rather than solving the constraint, we build momenta that are *manifestly* rank-≤ 4 for every value of the slice parameter, so Proposition 1 applies for free.

2 Massless spinor-helicity preliminaries

A massless momentum is written as a bispinor product

$$p^{\alpha\dot{\alpha}} = \lambda^{\alpha}\tilde{\lambda}^{\dot{\alpha}}, \tag{2}$$

with $\lambda, \tilde{\lambda} \in \mathbb{C}^2$. Since $p^{\alpha\dot{\alpha}}$ is manifestly rank one, $\det p = p^2 = 0$ automatically – masslessness is not imposed, it is built in. Antisymmetric spinor brackets are

$$\langle i\, j\rangle \coloneqq \lambda_i^1\lambda_j^2 - \lambda_i^2\lambda_j^1, \qquad [i\, j] \coloneqq \tilde{\lambda}_i^1\tilde{\lambda}_j^2 - \tilde{\lambda}_i^2\tilde{\lambda}_j^1, \tag{3}$$

and the basic dictionary entry we will use repeatedly is

$$2\, p_i{\cdot}p_j \;=\; \langle i\, j\rangle\, [j\, i]. \tag{4}$$

3 The all-line holomorphic shift

Pick a fixed reference spinor η and numbers $c_1, \ldots, c_n$, and deform every holomorphic spinor by a common multiple of η while leaving every anti-holomorphic spinor untouched:

$$\lambda_i(t) \;=\; \lambda_i + t\, c_i\, \eta, \qquad \tilde{\lambda}_i(t) \;=\; \tilde{\lambda}_i. \tag{5}$$

This is the shift used (for example) in [1, 2] to build univariate slices for functional reconstruction; it is a different object from the classic two-particle BCFW recursion shift (which deforms one leg's λ and another leg's $\tilde{\lambda}$ oppositely), but shares its spirit: a holomorphic deformation that preserves on-shellness and momentum conservation exactly.

On-shellness. Trivial: $\tilde{\lambda}_i$ never moves, so $p_i(t) = \lambda_i(t)\tilde{\lambda}_i(t)$ stays rank one, hence null, for every t.

Momentum conservation. $\sum_i p_i(t) = \sum_i p_i + t\sum_i c_i(\eta\tilde{\lambda}_i)$ as a bispinor; the $O(t)$ term vanishes iff

$$\sum_i c_i\, \tilde{\lambda}_i \;=\; 0, \tag{6}$$

one linear (2-component) equation in the c_i. Any solution works; with n unknowns and 2 constraints there are $n-2$ free parameters left over, which is where the genericity of the slice comes from.

Linearity of every invariant. Expand the shifted angle bracket:

$$\langle i\, j\rangle(t) = \langle i\, j\rangle + t\big(c_i\langle\eta\, j\rangle + c_j\langle i\,\eta\rangle\big) + t^2 c_i c_j\langle\eta\,\eta\rangle. \tag{7}$$

The t^2 term vanishes identically because $\langle\eta\,\eta\rangle = 0$ for any spinor (antisymmetry) – not a special property of this configuration, true for *any* choice of η. So every angle bracket is *exactly* linear in t; since $[j\, i]$ never moves, every $2p_i \cdot p_j$ in (4), and hence every Mandelstam invariant (a sum of such terms), is linear in t.

The Gram constraint. For every t, the $p_i(t)$ are honest null, momentum-conserving 4-vectors – nothing about t changes that. Proposition 1 therefore applies for *every* t simultaneously: $G(p_{i_1}(t), \ldots, p_{i_5}(t)) \equiv 0$ as a polynomial in t, not just at $t = 0$. This is the entire point of the construction: the constraint is never solved, it is a free corollary of building genuinely 4D momenta.

3.1 Worked example: a massless hexagon

Take $n = 6$ massless momenta. Four lines are chosen freely,

$$\begin{aligned} \lambda_1 &= (1,0), \quad & \tilde{\lambda}_1 &= (1,1), \\ \lambda_2 &= (0,1), \quad & \tilde{\lambda}_2 &= (2,1), \\ \lambda_3 &= (1,1), \quad & \tilde{\lambda}_3 &= (1,-1), \\ \lambda_4 &= (1,-1), \quad & \tilde{\lambda}_4 &= (1,2), \end{aligned} \tag{8}$$

and lines 5,6 are then *not* free: for six massless momenta to satisfy momentum conservation at all, $p_5 + p_6$ must equal the specific (generically non-null) bispinor $Q = -(p_1 + p_2 + p_3 + p_4)$, which only has a rank-one (null+null) decomposition for special λ_5. Concretely: pick $\lambda_5 = (2,1)$ freely, then $\det(Q - \lambda_5 \otimes \tilde{\lambda}_5) = 0$ is one linear equation fixing $\tilde{\lambda}_5$ (this is the same type of single linear solve as (6); see Remark 1). Solving it and reading p_6 off the resulting rank-one remainder gives

$$\lambda_5 = (2,1),\ \tilde{\lambda}_5 = (1,-16), \qquad \lambda_6 = (-5,-3),\ \tilde{\lambda}_6 = (1,-6). \tag{9}$$

One checks directly that this point already satisfies $G(p_1, p_2, p_3, p_4, p_5) = 0$ – it has to, by Proposition 1, since these are six genuine on-shell momenta summing to zero.

Choosing the reference spinor $\eta = (1,2)$ and solving (6) (after freely fixing $c_2 = c_4 = 0$ for simplicity, then solving c_5, c_6; an overall rescaling is used to clear denominators) gives

$$(c_1, c_2, c_3, c_4, c_5, c_6) = (5,\ 0,\ 5,\ 0,\ 6,\ -16). \tag{10}$$

Evaluating (4) along the shift (5) with these numbers gives, exactly:

$$\begin{aligned} s_{12}(t) &= 1 + 5t, & s_{23}(t) &= -3 - 15t, & s_{34}(t) &= 6 + 45t, \\ s_{45}(t) &= 54 + 324t, & s_{56}(t) &= 10 + 60t, & s_{16}(t) &= -21 + 21t, \\ s_{123}(t) &= 2 + 10t, & s_{234}(t) &= 9 + 30t, & s_{345}(t) &= 30 + 99t, \end{aligned} \tag{11}$$

and direct expansion of the 5×5 determinant confirms

$$G\big(p_1(t), p_2(t), p_3(t), p_4(t), p_5(t)\big) \;=\; 0 \qquad \text{identically in } t. \tag{12}$$

Every invariant is manifestly degree one in t, and the constraint never had to be imposed by hand.

Remark 1 (The "last two legs" completion)**.** Generating the base point itself (choosing $\lambda_5, \tilde{\lambda}_5, \lambda_6, \tilde{\lambda}_6$ so that $p_5 + p_6 = Q$ with both null) uses the same elementary trick throughout this note: for a 2×2 bispinor M and a fixed v, $\det(M - v \otimes w)$ is *linear*, not quadratic, in w (a standard adjugate identity), so fixing v and solving $\det(M - v \otimes w) = 0$ for w is always a single linear equation. This is reused twice below for the massive base point.

4 What changes for massive legs

Now suppose legs 5 and 6 are massive, $p_5^2 = p_6^2 = m_t^2 \neq 0$ (e.g. a top/antitop pair), with legs 1–4 still massless – exactly the external kinematics of $t\bar{t}b\bar{b}$-type processes. A massive momentum has no single null $(\lambda, \tilde{\lambda})$ pair, so (5) cannot be applied to it directly. The fix changes remarkably little.

4.1 Split each massive leg into two null constituents

Write

$$p_5 = q_{5a} + q_{5b}, \qquad p_6 = q_{6a} + q_{6b}, \tag{13}$$

with all four q's individually null. This is not a trick specific to this computation: it is exactly the two-component little-group decomposition of a massive momentum, $p^{\alpha\dot\alpha} = \lambda^{I\alpha}\tilde\lambda_I^{\dot\alpha}$ ($I = 1, 2$), with q_{5a}, q_{5b} the two little-group components. Using (4),

$$m_t^2 = p_5^2 = 2\, q_{5a}\cdot q_{5b} = \langle 5a\, 5b\rangle[5b\, 5a], \tag{14}$$

so *the mass is just one more spinor bracket.* Generating the $t = 0$ base point now means: pick $\lambda_{5a}, \tilde\lambda_{5a}, \lambda_{5b}$ freely, then (14) is one linear equation fixing $\tilde\lambda_{5b}$ (same mechanism as Remark 1); likewise for leg 6; then the remaining two legs (1,2 free, 3 free + 4 completed, say) are fixed by overall momentum conservation exactly as before, now summing over all eight null constituents $\{1, 2, 3, 4, 5a, 5b, 6a, 6b\}$ instead of six.

4.2 The shift, and one new ingredient

Apply the *same* all-line shift (5) to all eight constituent spinors. Nothing in the on-shellness or momentum-conservation arguments of Section 3 changes – (6) simply has eight terms instead of six. What is new is that we must decide how the mass behaves along the slice, since (14) will itself move under the shift:

$$m_t^2(t) = \langle 5a\, 5b\rangle(t)\,[5b\, 5a] = \Big[\langle 5a\, 5b\rangle + t\big(c_{5a}\langle\eta\, 5b\rangle + c_{5b}\langle 5a\,\eta\rangle\big)\Big][5b\, 5a], \tag{15}$$

linear in t by exactly the same bracket argument as before ($[5b\, 5a]$ never moves). Two options:

- **Hold m_t^2 fixed.** Demand the $O(t)$ term above vanishes:

  $$c_{5a}\langle\eta\, 5b\rangle + c_{5b}\langle 5a\,\eta\rangle = 0, \qquad c_{6a}\langle\eta\, 6b\rangle + c_{6b}\langle 6a\,\eta\rangle = 0, \tag{16}$$

  one linear constraint per massive leg (2 total). Useful if you want several slices at different fixed physical masses.

- **Let m_t^2 run, shared between the top and antitop.** Since $p_5^2 = p_6^2$ must hold identically along a *physical* slice (same particle, same mass), tie the two slopes together instead of killing them:

  $$[5b\, 5a]\big(c_{5a}\langle\eta\, 5b\rangle + c_{5b}\langle 5a\,\eta\rangle\big) \;=\; [6b\, 6a]\big(c_{6a}\langle\eta\, 6b\rangle + c_{6b}\langle 6a\,\eta\rangle\big), \tag{17}$$

  one linear constraint total. Now m_t^2 becomes a genuine extra linear slice direction, exactly on the same footing as $s_{12}, \ldots, s_{56}$ – which is the more useful choice in practice, since it lets a single line probe the mass-dependence together with the Mandelstam-dependence.

Either way: 8 unknowns c_k, 2 (momentum conservation) + 1 or 2 (mass handling) linear constraints, leaving 4–5 free parameters – still plenty of room for a generic slice.

4.3 Why the Gram constraint is still automatic

Nothing here touches Proposition 1. The eight constituents $q_k(t)$ are still honest null bispinors for every t (only λ's moved), and (6) (now an eight-term sum) still enforces momentum conservation exactly for every t. So $p_1(t), \ldots, p_4(t), p_5(t){=}q_{5a}(t){+}q_{5b}(t), p_6(t)$ remain six genuine 4D momenta summing to zero for every t, regardless of what $m_t^2(t)$ does – the Gram determinant of any five of them vanishes identically, for exactly the same linear-algebra reason as in the massless case. Masses never entered Proposition 1's proof, so they cannot break it here either.

4.4 Worked example: extending the hexagon to $t\bar{t}b\bar{b}$-type kinematics

Keep legs 1,2 as before, and replace legs 3–6 by: a free leg 3, a completed leg 4, and two massive legs 5,6 with $m_t^2 = 4$, each split into null constituents. Choosing

$$\begin{aligned} &\lambda_{5a} = (1,1), \quad &&\tilde{\lambda}_{5a} = (1,-1), \quad &&\lambda_{5b} = (1,2), \quad &&\tilde{\lambda}_{5b} = (1,-5), \\ &\lambda_{6a} = (1,-1), \quad &&\tilde{\lambda}_{6a} = (1,1), \quad &&\lambda_{6b} = (2,1), \quad &&\tilde{\lambda}_{6b} = \left(1,-\tfrac{1}{3}\right), \end{aligned} \tag{18}$$

one checks $\langle 5a\,5b\rangle[5b\,5a] = \langle 6a\,6b\rangle[6b\,6a] = 4 = m_t^2$, exactly as designed. The same "last-two-legs" completion (Remark 1) then fixes $\lambda_3 = (1,1)$, $\tilde{\lambda}_3 = (1, \frac{154}{3})$ and $\lambda_4 = (-7,-6)$, $\tilde{\lambda}_4 = (1, \frac{20}{3})$ so that overall momentum conservation holds exactly. At this base point one finds, directly from the definitions,

$$m_t^2 = 4, \quad s_{12} = 1,\ s_{23} = \tfrac{305}{3},\ s_{34} = \tfrac{134}{3},\ s_{45} = -97,\ s_{56} = \tfrac{134}{3},\ s_{16} = \tfrac{16}{3},\ s_{123} = \tfrac{157}{3},\ s_{234} = 60,\ s_{345} = 4, \tag{19}$$

and, as required, $G(p_1,p_2,p_3,p_4,p_5) = 0$ (the fractions are just an artifact of the small integers chosen for the constituent spinors; nothing here needs them to be especially pretty).

Choosing $\eta = (1,-1)$ and the "shared running mass" option, solving the resulting 3×3 linear system (after freely fixing $c_1 = c_4 = c_{5a} = 1$, $c_2 = c_3 = 0$, then clearing a common denominator of 22) gives

$$(c_1, c_2, c_3, c_4, c_{5a}, c_{5b}, c_{6a}, c_{6b}) = (22,\ 0,\ 0,\ 22,\ 22,\ 35,\ -4,\ -97). \tag{20}$$

Evaluating every invariant along this shift gives, exactly:

$$\begin{aligned} &m_t^2(t) = 4 - 16t \ (= p_6^2(t) \text{ as well}), \quad &&s_{12}(t) = 1 + 22t, \\ &s_{23}(t) = \tfrac{305}{3} \ \text{(const.)}, \quad &&s_{34}(t) = \tfrac{134}{3} - \tfrac{5896}{3}t, \\ &s_{45}(t) = -97 + \tfrac{25777}{3}t, \quad &&s_{56}(t) = \tfrac{134}{3} - \tfrac{6424}{3}t, \\ &s_{16}(t) = \tfrac{16}{3} + \tfrac{604}{3}t, \quad &&s_{123}(t) = \tfrac{157}{3} - \tfrac{6578}{3}t, \\ &s_{234}(t) = 60 - 1694t, \quad &&s_{345}(t) = 4 + 381t, \end{aligned} \tag{21}$$

and again $G(p_1(t), p_2(t), p_3(t), p_4(t), p_5(t)) \equiv 0$. (Here s_{23} happens to be constant only because we deliberately set $c_2 = c_3 = 0$ to keep the arithmetic short; for a random choice of free parameters every invariant, m_t^2 included, has a nonzero slope.) Every invariant, *including the mass*, is exactly degree one in t, and the Gram constraint was never solved – only ever a consequence of using genuine on-shell, momentum-conserving spinor data.

5 Summary: massless versus massive

	Massless (§3.1)	**Massive (this note)**
Lines shifted	n external legs	n_{massless} external legs + 2 null constituents per massive leg
Momentum conservation	1 linear constraint on c_i ((6))	same, just more terms
Mass control	—	1 extra linear constraint per fixed mass, or 1 shared constraint to let a tied mass run
Invariant linearity in t	automatic (bracket identity)	automatic, by the same identity, now including m_t^2 itself
Gram constraint	automatic (Prop. 1)	automatic, unchanged – masses never enter the proof
Square roots	none	none

In short: nothing about the core argument changes. The only new step is representing each massive leg as a sum of two null constituents (equivalently: using genuine massive little-group spinors) so that its mass becomes just another spinor bracket – at which point it is controlled by exactly the same kind of single linear equation used everywhere else in the construction. This is precisely what `MassiveBCFWSlice.wl` implements, with finite-field arithmetic in place of exact rationals and a free toggle (`FixMasses`) between the two mass-handling options above.

References

[1] S. Abreu, P. F. Monni, B. Page, J. Usovitsch, *Planar Six-Point Feynman Integrals for Four-Dimensional Gauge Theories*, arXiv:2412.19884.

[2] S. Abreu, F. Febres Cordero, H. Ita, M. Klinkert, B. Page, V. Sotnikov, *Leading-Color Two-Loop Amplitudes for Four Partons and a W Boson in QCD*, JHEP **04** (2022) 042, arXiv:2110.07541.

[3] M. Becchetti et al., *Two-loop QCD amplitudes for $t\bar{t}W$ production at the LHC in the leading-colour approximation*, arXiv:2608.03746.